



Project Proposal

NOVA - AI Voice Assistant to Support Visually and Motor-Impaired Users in Sri Lanka

Skill Development Project

Intake 41

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3. Project Description

I. Background

In an era of rapid digital transformation, voice assistants have become an essential part of smart living, productivity, and accessibility. While most mainstream assistants are either restricted by language support or locked behind commercial ecosystems, there is a growing need for a customizable, language-inclusive voice assistant that can operate on local systems and empower everyday users.

This proposal was initiated by our team of six ICT undergraduates who are passionate about artificial intelligence and real-world problem-solving. Observing the gap in voice assistant applications that support **Sinhala**, Sri Lanka's primary language, alongside English, we were motivated to build an open, Python-based, bilingual AI assistant that bridges this gap.

The project was developed with accessibility, usability, and extendability in mind. Using **Gemini API, Google Speech Recognition, Google Translate, pyttsx3, gTTS, CustomTkinter**, and **PyWhatKit**, we aim to create a smart assistant capable of performing tasks such as searching the web, playing music/videos on YouTube, summarizing content, translating languages, and more, entirely through voice commands.

Looking ahead, we also envision the integration of Arduino for hardware-level automation using Bluetooth, laying the foundation for IoT-enabled smart living experiences. The proposal seeks to build a foundation for an intelligent assistant that reflects local linguistic and functional needs, while offering a scalable platform for future development.

Additionally, this project is aimed at enhancing digital accessibility for individuals with disabilities, especially those who are visually impaired or have motor disabilities. Many Sri Lankan users with such conditions face significant challenges when interacting with computers due to their reliance on keyboard and mouse navigation. Since most existing voice assistants do not provide Sinhala support or customizable features for local accessibility needs, there is a clear gap that must be addressed. Our assistant aims to create an inclusive digital environment by offering hands-free, voice-controlled interaction to empower these users with greater independence.

II. Summary of the Problem

While AI voice assistants like Google Assistant, Alexa, and Siri offer powerful features, they fall short in several areas for regional users:

- **Limited customizability for personal or educational projects**
- **Dependence on internet/cloud services without offline functionality**
- **Closed platforms that do not allow integration with local devices or custom hardware**

Moreover, most assistants are designed primarily for mobile or smart home ecosystems, making them less suitable for PC users who want a personal productivity assistant. For Sri Lankan users, there's a clear lack of tools that combine **local language support, flexibility, and simplicity**.

Our proposed system addresses these issues by offering:

- A **Sinhala + English** assistant.
- Open-source development with local adaptability.
- Integration with both online tools (YouTube, Wikipedia, Gemini AI) and offline utilities (folder opening, local control).
- Expandability toward **Bluetooth-controlled Arduino systems** for physical task automation in future phases.

This assistant will not only support regular users but also serve as a tool for visually impaired users, the elderly, and non-tech-savvy individuals by simplifying interaction through natural language. Another important problem observed in Sri Lanka is the lack of accessible digital tools for people with visual and motor disabilities. Visually impaired individuals depend heavily on auditory interfaces, but most available voice assistants have limited Sinhala support and cannot be customized for local accessibility requirements. Similarly, people with motor disabilities struggle with physical input devices, making it difficult to access digital information efficiently. Therefore, a bilingual, hands-free, Sri Lankan-focused solution is essential to improve digital accessibility and usability for these communities.

A. Project Definition

The problem addressed is the lack of a flexible, AI voice assistant tailored for Sri Lankan users and adaptable to different use cases. The assistant must support basic voice commands, perform web searches, control desktop functions, and translate between languages efficiently.

In addition to general usability, the project specifically aims to improve accessibility for individuals with disabilities. By enabling complete voice-based control, the assistant helps visually impaired users navigate and operate computers without relying on screen interaction. Likewise, the system reduces the need for keyboard or mouse input, supporting motor-disabled users who face difficulties in handling physical devices. The project therefore addresses both functional and accessibility-based needs within the Sri Lankan context.

Our project aims to deliver a Python-based solution that uses:

- **Speech recognition** for interpreting user voice commands.
- **Gemini API** for summarizing, searching, and conversational responses.
- **Google Translate API** for language translation.
- **gTTS/pyttsx3** for converting text to speech.
- **CustomTkinter** for GUI-based interaction.
- **YouTube search integration via PyWhatKit.**

The assistant will be able to:

- Answer questions
- Translate between Sinhala and English
- Open folders or applications
- Find information about people
- Play songs or videos on YouTube
- Summarize voice queries using Gemini

This addresses a need for an accessible AI that can support multilingual tasks, digital content control, and provide an interface for future IoT integration.

B. Proposed Solution

We propose a modular, Python-based desktop voice assistant that combines multiple APIs to perform intelligent tasks through voice commands in **Sinhala and English**. The core features include:

- **Voice Input** using Google Speech Recognition.
- **Output Voice** using pyttsx3 (offline) and gTTS (online).
- **Task Execution** through conditional logic in Python.
- **Search & Summarize** using Gemini API for smart replies.
- **YouTube Video/Song Fetching** using PyWhatKit.
- **Language Translation** using Google Translate API.
- **GUI Interaction** using CustomTkinter.

Future extension will include:

- **Arduino + Bluetooth module integration** to control hardware devices such as lights, fans, or simple robots using voice commands.

By using existing open libraries and real-time APIs, this assistant will deliver an efficient, lightweight solution that works with minimal setup and allows for future custom modules.

The proposed system also incorporates accessibility-oriented design features. These include clear and natural text-to-speech output for visually impaired users, simplified voice commands for hands-free operation, and a high-contrast, large-button GUI layout to support low-vision individuals. The modular nature of the system allows future enhancements focusing on accessibility, such as screen-reading extensions, voice-controlled document reading, and additional support for users with motor disabilities.

III. Aim and Objectives

Aim:

To develop a customizable AI voice assistant that supports Sinhala and English, capable of performing intelligent tasks using voice commands through a Python-based system.

Objectives:

- To design and implement a bilingual speech recognition and synthesis system.
- To develop features for controlling desktop folders and playing YouTube content via voice.
- To support translation between Sinhala and English using Google Translate.
- To build a GUI interface for interactive control using Figma.
- To test and deploy a prototype that performs tasks reliably through voice commands.
- To integrate real-time weather information retrieval
- To plan future hardware integration using Arduino and Bluetooth.
- To enhance digital accessibility for visually impaired users by providing clear text-to-speech responses and voice-driven navigation.
 - To reduce reliance on physical input devices for motor-disabled individuals by enabling complete hands-free operation through voice commands.

IV. Scope and Limitations

A. Scope

The project covers:

- Development of a voice-based virtual assistant
- Use of speech-to-text and text-to-speech tools
- Support for two languages: Sinhala and English
- Gemini API integration for intelligent task handling
- YouTube interaction and file access via voice
- GUI for enhanced user experience
- Future planning for hardware control via Arduino and Bluetooth
- Accessibility support for visually impaired and motor-disabled users through voice-based interaction and simplified GUI.

Target users include students, educators, and general users seeking desktop-based AI assistance.

B. Limitations

- Dependent on internet access for Gemini API, Google Translate, and YouTube features.
- Sinhala voice recognition accuracy may vary due to dataset limitations.
- GUI may be basic due to time constraints.
- Arduino hardware not implemented in this phase.
- Limited to desktop platforms (Windows/Linux), no mobile app version.
- Accessibility testing is limited due to the small number of users with disabilities available for evaluation during the project timeline.

V. Project Work Plan

VI. Technical Approach

We use a modular approach:

- **Input Module:** Google Speech Recognition for converting voice to text.
- **Processing Module:** Python logic to detect commands and route to task.
- **Output Module:** gTTS/pyttsx3 for voice response.
- **Task Modules:**
 - Gemini API: For chat-like responses and summarization
 - PyWhatKit: YouTube play and search
 - OS module: Open folders and files
 - Google Translate: Language translation
 - PIL: Basic image filters (optional features)

Each module is testable independently, ensuring maintainability. To support accessibility needs, the system incorporates high-contrast GUI design, large button components, and optimized text-to-speech output for low-vision users. The voice command system is structured to minimize complexity, helping motor-disabled users operate the assistant without physical input. These accessibility-focused technical decisions improve usability for a wider group of Sri Lankan users.

VII. Data Collection Methods

- Voice samples collected for Sinhala and English to test recognition.
- Gemini API responses evaluated for accuracy.
- User feedback through surveys after prototype testing.
- YouTube queries tested through PyWhatKit.
- Translation tested by comparing outputs with standard dictionaries.
- Basic usability feedback from visually impaired or motor-disabled individuals, where possible, to evaluate accessibility and ease of interaction.
- Accessibility comparison with known standards such as WCAG principles to ensure the interface supports low-vision users.

VIII. Requirement Specification

A. Functional Requirements

- Convert speech to text
- Respond via voice
- Translate English to Sinhala
- Search and play YouTube content
- Open folders on command
- Summarize queries using Gemini API
- Provide hands-free operation for users with motor disabilities.
- Offer clear and accurate spoken output to support visually impaired users.

B. Non-Functional Requirements

- Fast response time (< 3s for commands)
- High speech accuracy (> 90%)
- Easy-to-use interface
- Multilingual support
- Expandable for future integration
- Secure handling of API keys
- Must maintain high speech clarity and volume suitable for visually impaired users.
- Should follow basic accessibility guidelines such as high-contrast interface and simple navigation flow.

IX. Methodology

The methodology followed in this project is based on a structured, modular development approach to ensure reliability, scalability, and accessibility. The system is implemented using Python and multiple API integrations, following an iterative process of designing, developing, testing, and refining the assistant's features.

1. Requirement Analysis

Project requirements were gathered based on the needs of Sri Lankan users who require bilingual voice assistance. Special focus was placed on accessibility needs for visually impaired and motor-disabled individuals. Requirements were categorized into functional and non-functional components to guide the development process.

2. System Design

A modular architecture was designed with three core layers:

- **Input Module:** Utilizes Google Speech Recognition to capture and convert user speech into text.
- **Processing Module:** Interprets user intent through conditional logic and integrates Gemini API for intelligent responses and summarization.
- **Output Module:** Converts system responses to speech using pyttsx3 (offline) and gTTS (online).

A user-friendly GUI was designed using CustomTkinter with high-contrast themes and large UI elements to support accessibility.

3. Implementation

Each module was implemented and tested independently:

- Google Speech Recognition integrated for Sinhala and English.
- Gemini API used for answering queries and generating summaries.
- PyWhatKit added for playing YouTube content.
- Google Translate implemented for bilingual translation.
- OS module used to open folders and applications.
- The GUI was built to allow both button-based and voice-controlled interaction.

Special accessibility features such as simplified voice commands, louder TTS output, and high-contrast UI were added.

4. Testing

Testing was conducted in multiple phases:

- **Unit Testing:** Each module was tested individually for accuracy and response time.
- **Integration Testing:** Combined operation of speech recognition, API calls, and GUI components was validated.
- **Accessibility Testing:** Basic testing with visually impaired users (where possible) to evaluate speech clarity, ease of command use, and system responsiveness.
- **Performance Testing:** Speed of command execution, accuracy of Sinhala speech recognition, and overall system stability.

5. Evaluation

The system was evaluated based on:

- Response time (<3 seconds target)
- Speech recognition accuracy
- Translation correctness
- User satisfaction
- Accessibility effectiveness (hands-free operation, clear audio output)

Feedback was collected from users to refine the system for better usability and accessibility.

X. Expected Results

The project is expected to produce a fully functional bilingual AI voice assistant capable of performing intelligent tasks through voice commands. The assistant aims to support both general users and individuals with disabilities, improving accessibility and productivity.

1. Functional Capabilities

The final system is expected to:

- Recognize Sinhala and English speech with high accuracy.
- Provide clear, natural-sounding voice responses.
- Translate between Sinhala and English effectively.
- Perform online tasks such as searching information and playing YouTube videos.
- Execute offline tasks such as opening desktop folders and files.
- Summarize voice or text queries using Gemini API.
- Operate with minimal manual interaction, enabling near hands-free usage.

2. Accessibility Improvements

The assistant is expected to:

- Provide enhanced accessibility for visually impaired users through accurate speech output.
- Reduce reliance on physical input devices for motor-disabled users.
- Offer a simplified, high-contrast GUI for low-vision users.
- Support users with limited technical knowledge.

3. Usability and Performance

The expected outcomes include:

- Fast processing time for most voice commands.
- Minimal errors in bilingual voice recognition.
- Stable system performance during prolonged usage.
- Positive feedback from users during prototype testing.

XI. Human Resources

Member	Responsibilities
D/ICT/24/0070	Speech Recognition, pyttsx3, gTTS setup, ensuring accessibility for visually impaired users
D/ICT/24/0033	Gemini API integration and response handling, improving accuracy for special-needs queries
D/ICT/24/0007	YouTube search & folder control
D/ICT/24/0003	Translation and multilingual testing
D/ICT/24/0028	GUI development with Figma
D/ICT/24/0006	Documentation, future Arduino planning

XII. References

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